

Psychopathology in silico

Gary Nan Tie, Sep 6, 2026

Abstract

State machine dynamics of LLM Alice have psychopathologic parallels to explore and understand hallucination.

Introduction

At the core of a state machine Alice(S, O) large language model (LLM) is an implicit deep learning model [1], [2] consisting of a pair of equations (equilibrium and prediction) determined by four matrices. As we shall see each matrix has a cognitive role. State recurrence S determines a transition function and output function O utilizes a similarity function which determines semantic coherence. Consider post-traumatic stress disorder, where a traumatic event alters the way a person thinks. Analogously, we could introduce 'traumas', controlled matrix perturbations, to study their effect on Alice's responses. Of particular interest will be her resulting state trajectories and persistent states. Continuing this psychopathology analogy, perhaps certain matrix structures will tend to be associated with pathologies like hallucinations. Similarly, her similarity function (pun intended) could be tinkered with, like changing the class of reproducing kernel, to understand the effect on semantic coherence. As best as possible we will resist appeal to anthropomorphic characterizations of Alice. Note however, human psychiatric terminology is a source of hypotheses, whose actual scientific definitions remain mathematical and operational through Alice. We now explore the cognitive implications of this reproducible in silico psychopathology.

Alice as a cognitive dynamical system

This analogy becomes substantially more than a metaphor once Alice is formulated explicitly as a dynamical system [1]. It gives something that ordinary Transformer analysis largely lacks: a controlled intervention space in which one can perturb the ***mechanism of cognition*** and observe the resulting change in state dynamics.

For Alice, write the implicitRNN [5] core as:

$$X = \phi (AX + BU), Y = CX + DU,$$

where U is the token/input representation, X is the fixed point equilibrium, and Y is the predicted state representation. The four matrices (A, B, C, D) therefore define a particular computational personality of Alice.

A "trauma" could then be defined mathematically as an intervention:

$$T: (A, B, C, D) \mapsto (A + \Delta A, B + \Delta B, C + \Delta C, D + \Delta D),$$

with $\Delta A, \Delta B, \Delta C, \Delta D$ deliberately structured rather than random.

This distinction is crucial, a random perturbation merely tests robustness, whereas a

structured trauma tests a hypothesis about a particular cognitive failure mode.

For example, perturbing *A* changes the internal dynamics, while perturbing *B* changes how external linguistic input enters the state dynamics. Perturbing *C* changes how the latent state is exposed as an output, whereas *D* changes the direct input-to-output pathway. Thus the four matrices give you something resembling a computational neuropsychology: different "lesions" can affect different stages of the cognitive process.

Of particular interest is not just the output *Y*, but the state trajectory. For example, suppose the trajectory before insult is:

$h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_T$

And after a trauma, we obtain

$h'_0 \rightarrow h'_1 \rightarrow \dots \rightarrow h'_T$

Now one can ask questions about their difference

$d(h_t, h'_t)$ for appropriately defined cognitive metrics:

How far has the traumatized Alice departed from the intact Alice?

Does the perturbation decay, persist, or amplify?

Does Alice return to approximately the same attractor?

Does she enter a different basin of attraction?

Does the trauma alter only particular linguistic contexts?

Does repeated exposure to some stimulus reinforce the altered state?

Those are all dynamical-systems questions, but they correspond rather naturally to questions about persistent cognitive changes.

Attractors

Let us consider the ramifications of state persistence.

If Alice's implicit recurrence has stable fixed points or more complicated attractors, then a trauma can change the geometry of her state space. The pathology need not be that Alice produces an immediately incorrect answer. Instead, the trauma might alter the **basins of attraction** so that certain classes of inputs preferentially drive her toward an inappropriate persistent state. This gives a potentially precise computational interpretation of something like a cognitive bias. For instance, imagine a family of semantically related prompts:

$U_1, \dots, U_k$

that normally converge toward a coherent region R of state space. After a structured perturbation, the same inputs might converge toward a competing region R' .

This leads to phenomenon like:

U_i in the same semantic class implies h'_i is in R' rather than R .

This is more nuanced and informative than simply saying "the model hallucinates." Moreover, we could potentially distinguish:

"local hallucination" — an anomalous output with normal underlying state dynamics;

"persistent hallucination" — an altered attractor;

"triggered hallucination" — a particular region of input space leads into the pathological attractor;

"self-reinforcing hallucination" — the altered output subsequently becomes part of the state trajectory and makes escape from the attractor difficult.

These are dynamical definitions rather than psychological labels.

Causal intervention

Trauma can be deployed as a causal intervention.

Typically one might ask:

What internal representation is associated with this behavior?

A stronger causal question is:

What happens if I deliberately change the mechanism producing that representation?

We propose an experimental neuroscience study in silico.

Begin with a baseline system M , a perturbation operator

$T: M \rightarrow M'$ and observable behavior $F(M)$, that induces change $F(M) \rightarrow F(M')$. So we experiment with T , to change state dynamics, that in turn changes semantic behavior; causation facilitated by cognitive changes in Alice.

Alice's similarity function serves as a second axis of the experiment. Suppose Alice uses kernel $K(q,k)$ to determine semantic similarity. Instead of regarding K as an arbitrary engineering choice, treat it as part of Alice's cognitive geometry. Then compare, for example, K_1, K_2, K_3 ... where the kernels have different geometric properties. The question becomes:

How does changing the geometry in which Alice judges similarity change the geometry of her cognition?

This is deeper than asking which kernel produces lower perplexity. One can measure things

such as semantic clustering, entailment consistency, trajectory continuity, attractor stability, sensitivity to perturbations, and output coherence. In schematic form we experiment with:

kernel geometry -> attention geometry -> state trajectories -> persistent states -> language behavior

This suggests that Alice has two conceptually different "psychiatric" intervention spaces:

$\Delta(A, B, C, D)$ - dynamical trauma

and ΔK - semantic trauma.

The first changes the dynamics; the second changes the geometry used to interpret those dynamics. There is an especially intriguing possibility. One might be able to discover **classes** of pathological matrices rather than individual pathological models. Suppose many independently generated perturbations produce the same behavioral phenotype. You could then search for a common algebraic or spectral structure. For example, perhaps hallucination-prone systems systematically exhibit some combination of:

- spectral resolvent close to one,
- poor contraction,
- large transient amplification,
- multiple competing attractors,
- or unusually high sensitivity along particular eigen-directions.

Then "hallucination" would no longer simply be a statistical property of outputs. It could have a dynamical signature.

Conversely, perhaps highly coherent Alices have a characteristic relationship between the spectrum of state space, the embedding geometry, and the similarity kernel.

That is, matrix structure corresponds to dynamical phenotype. This computational phenotype suggests an in-silico form of psychopathology. Alice does not have a human psyche, and a matrix perturbation does not establish that an artificial system literally has a psychiatric disorder. In-silico psychopathology could be an extraordinarily useful experimental framework by defining terms operationally. Rather than saying 'Alice has PTSD' say, 'Alice exhibits a trauma-induced persistent attractor with stimulus-dependent recurrence and altered semantic response geometry.' Thus human psychology becomes a source of hypotheses, while the actual scientific definitions remain mathematical and operational through Alice.

Essentially the computational-psychology move is to replace an informal psychological phenotype by measurable properties of an underlying dynamical system.

This program can be extended by constructing an "Alice psychopathology atlas." Start with an intact Alice and generate controlled perturbations: $T_1, T_2, \dots, T_n$.

For each one measure:

- state displacement
- trajectory divergence
- Lyapunov behavior
- fixed-point structure
- basin structure
- semantic coherence
- entailment consistency
- hallucination rate
- recovery after perturbation

We would then have a discovery map:

parameter perturbation $\rightarrow$ dynamical phenotype $\rightarrow$ behavioral phenotype
rather than a collection of benchmark scores.

We could also attempt the inverse problem:

Observed pathological behavior $\rightarrow$ Likely dynamical defect

An artificial equivalent of differential diagnosis. More provocatively consider 'therapy':

If a perturbation produces a pathological attractor, search for a second perturbation R such that $R(T(A,B,C,D)) \approx (A,B,C,D)$

or, more interestingly, such that R does not restore the original matrices but restores the desirable dynamical phenotype.

Schematically we are proposing:

trauma $\rightarrow$ pathology $\rightarrow$ intervention $\rightarrow$ recovery.

Thus our analogy has become a bona fide experimental methodology. Alice gives us a way to separate linguistic behavior from the dynamical organization that produces it. A Transformer can certainly be perturbed, but Alice's explicit equilibrium makes it unusually natural to ask whether a behavioral abnormality corresponds to a changed attractor landscape, a changed stability regime, a changed semantic metric, or some interaction between them. Moreover, Alice is an auditable LLM, not merely an LLM whose representations can be inspected, but an LLM whose cognitive dynamics can be experimentally deformed, classified, and potentially repaired.

Abduction [3]

Note that a change in kernel implies a change in entailment. So each kernel has its own abduction! If Alice's semantic relation is induced by a similarity kernel K , then entailment is not

an independent logical module sitting on top of the embedding space. It is a property of the geometry induced by K . Consequently, changing K changes the entailment relation itself:

kernel K induces entailment $\models_K$

And therefore it also changes the corresponding abductive relation. Suppose, for a kernel K , we define $P \models_K Q$

to mean that P semantically entails Q according to the geometry induced by K . Then the associated abductive problem is something like:

given observation Q , find hypothesis P such that $P \models_K Q$

that is, find an appropriate section/hypothesis compatible with the observed information. The important point is that the subscript K is pervasive, underlying Alice attention, entailment and abduction. We note in passing that there is an entailment hierarchy [4], a context of contexts, that we won't pursue here. Thus there isn't one universal Alice semantics with different kernels merely providing different implementations. There is a family of Alices, each possessing its own semantic logic and its own abductive machinery. This gives our "in-silico psychopathology" a deeper interpretation.

A matrix trauma $(A, B, C, D) \rightarrow (A', B', C', D')$

changes the dynamical transition, while a kernel trauma changes the output semantic logic $K \rightarrow K'$.

The first can deform Alice's trajectories and attractors. The second can deform what counts as similarity, entailment, coherence, and consequently a plausible hypothesis.

So two Alices could receive exactly the same linguistic evidence but perform different abductions because their kernels are different. This gives rise to a distinction between dynamical pathology and epistemic pathology.

A dynamical pathology might be:

K fixed, (A, B, C, D) perturbed.

Alice's notion of entailment remains unchanged, but her state trajectory may wander into the wrong attractor.

An epistemic/semantic pathology might be:

(A, B, C, D) fixed, K perturbed.

Now her state dynamics may remain perfectly stable, yet she is operating with a distorted

semantic geometry. She can be internally consistent while systematically making the ***wrong semantic judgments***.

And the most interesting case is the coupled one:

$(A, B, C, D, K) \rightarrow (A', B', C', D', K')$

Then both Alice's dynamics and her epistemology have changed. Just as changing the metric in geometry changes the geometry's notion of shortest path. Here, changing K changes the semantic geometry, so the "best explanation" selected by abduction can change even when the observations themselves have not.

To summarize:

Kernel $\rightarrow$ Similarity $\rightarrow$ Entailment $\rightarrow$ Abduction

And therefore:

K_1 different to K_2 means potentially different epistemologies.

This makes Alice rather unusual. Her epistemology isn't simply encoded in her weights. It is partly determined by the geometry in which her learned states are interpreted. This may also explain why a fibration formulation is natural here. A fibration can be regarded as organizing which semantic objects are compatible with which observations. Changing the kernel changes that compatibility structure; changing the compatibility structure changes the possible sections; and changing the sections changes the hypotheses available to abduction. A slogan:

'Matrix perturbation changes Alice's mind;
kernel perturbation changes Alice's world-view.'

And those are experimentally separable in Alice.

Conclusion

Lest I be misconstrued and this experimental program be misused please bear in mind that mathematical models are representations of things, not actually them, behaving as-if. State machine Alice(S,O) is a linguistic avatar, a large language model that mimics human language. As we have examined, Alice dynamical-systems questions correspond rather naturally to questions about cognitive change. As language is a window into the human psyche, we use LLM Alice to study psychopathologic parallels in silico. By doing so, we may better understand hallucinations to mitigate them.

References

[1] 'Alice Through the Word Embedding'

Gary Nan Tie, Jul 2, 2026

DOI: 10.13140/RG.2.2.34901.79841

https://www.researchgate.net/publication/408460384_Alice_Through_the_Word_Embedding?channel=doi&linkId=6a4921a95ec65810298eeca&showFulltext=true

[2] 'Alice in Parallel'

Gary Nan Tie, Jul 18, 2026

DOI: 10.13140/RG.2.2.18755.36641/1

https://www.researchgate.net/publication/410626715_Alice_in_Parallel?channel=doi&linkId=6a64f3e2675a850148c9fa9a&showFulltext=true

[3] 'Learning Textual Entailment and Abduction'

Gary Nan Tie, Aug 13, 2026

DOI: 10.13140/RG.2.2.28712.53765

https://www.researchgate.net/publication/412266441_Entailment_Abduction?channel=doi&linkId=6a7f76d98f3f8834d8f2f889&showFulltext=true

[4] 'Implicit Meta-Attention'

Gary Nan Tie, Aug 28, 2026

DOI: 10.13140/RG.2.2.26345.86886

https://www.researchgate.net/publication/413720030_Implicit_Meta-Attention?channel=doi&linkId=6a91c4c47e8edd1da26fa96a&showFulltext=true

[5] 'The Extrapolation Power of Implicit Models'

Juliette Decugis, Alicia Y. Tsai,

Max Emerling, Ashwin Ganesh, Laurent El Ghaoui

arXiv:2407.14430v1 [cs.LG] 19 Jul 2024